Activity 1.1: Segments, Midpoints & Angle Basics

A

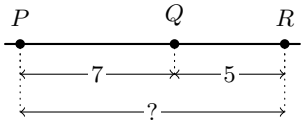
Name: _____ Partner's name: _____ Date: _____

Directions: Find the value or measure asked for. Show your work. Diagrams are accurate, but use math rather than a ruler or protractor.

Work alone, then compare after each problem. If answers differ, find the mistake together.

1

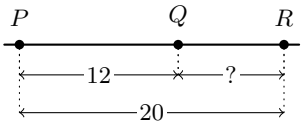
Points P, Q, R are collinear with Q between P and R. Find PR.



Answer: _____ Match? ☐ Yes ☐ No

2

Points P, Q, R are collinear with Q between P and R. Find QR.



Answer: _____ Match? ☐ Yes ☐ No

3

M is the midpoint of $\overline{AB}$. $AM = 2x+3$ and $MB = 4x-5$. Find AB.

Answer: _____ Match? ☐ Yes ☐ No

4

Find the midpoint of the segment with endpoints $(2, 3)$ and $(8, 9)$.

Answer: _____ Match? ☐ Yes ☐ No

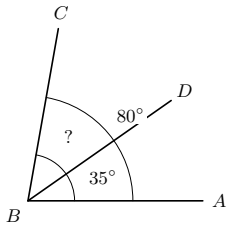
5

Find the distance between $(1, 2)$ and $(4, 6)$.

Answer: _____ Match? ☐ Yes ☐ No

6

D is inside $\angle ABC$. $m\angle ABC = 80°$ and $m\angle ABD = 35°$. Find $m\angle DBC$.



Answer: _____ Match? ☐ Yes ☐ No

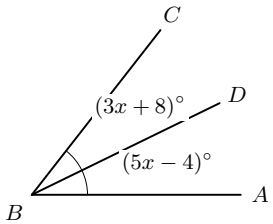
7

Two angles are complementary and measure $(3x + 5)°$ and $(2x + 10)°$. Find x.

Answer: _____ Match? ☐ Yes ☐ No

8

Ray BD bisects $\angle ABC$. Find $m\angle ABC$.



Answer: _____ Match? ☐ Yes ☐ No

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B

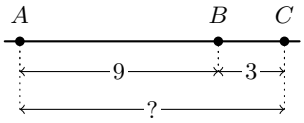
Name: _____ Partner's name: _____ Date: _____

Directions: Find the value or measure asked for. Show your work. Diagrams are accurate, but use math rather than a ruler or protractor.

Work alone, then compare after each problem. If answers differ, find the mistake together.

1

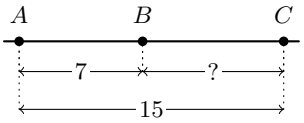
Points A, B, C are collinear with B between A and C. Find AC.



Answer: _____ Match? ☐ Yes ☐ No

2

Points A, B, C are collinear with B between A and C. Find BC.



Answer: _____ Match? ☐ Yes ☐ No

3

M is the midpoint of $\overline{CD}$. $CM = x + 6$ and $MD = 3x - 4$. Find CD.

Answer: _____ Match? ☐ Yes ☐ No

4

Find the midpoint of the segment with endpoints $(-1, 0)$ and $(11, 12)$.

Answer: _____ Match? ☐ Yes ☐ No

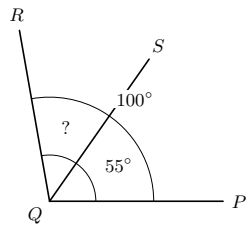
5

Find the distance between $(-2, 1)$ and $(1, 5)$.

Answer: _____ Match? ☐ Yes ☐ No

6

S is inside $\angle PQR$. $m\angle PQR = 100°$ and $m\angle PQS = 55°$. Find $m\angle SQR$.



Answer: _____ Match? ☐ Yes ☐ No

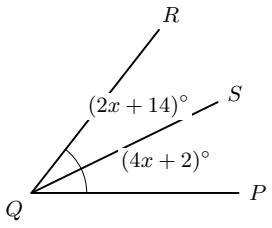
7

Two angles are supplementary and measure $(4x + 10)°$ and $(2x + 80)°$. Find x.

Answer: _____ Match? ☐ Yes ☐ No

8

Ray QS bisects $\angle PQR$. Find $m\angle PQR$.



Answer: _____ Match? ☐ Yes ☐ No